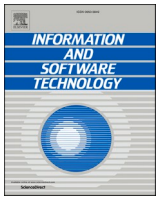
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Bridging the state-of-the-art and the state-of-the-practice of SaaS pricing: A multivocal literature review

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ABSTRACT

Context: Pricing is an essential element of software business strategy and tactics. Informed pricing decision-making requires the involvement of different stakeholders and comprehensive data analysis. Achieving both appears to be challenging, and pricing remains one of the most under-managed processes in the software business. Simultaneously, a coherent SaaS pricing body of knowledge and verified solutions to assist SaaS providers while designing and implementing pricing are missing.

Objective: There is a lack of integration among different research areas focused on SaaS pricing and, more importantly, between academia and industry. The primary aim of this paper is to clarify this misconception by classifying, thematically analyzing, and putting in correspondent academic state-of-the-art and industrial state-of-the-practice of SaaS pricing.

Method: A multivocal literature review (MLR) approach was used for the study, exploring both "white" literature as well as "grey" literature. The body of literature of 387 bibliography items was collected using a formal protocol. Of these, 57 were white literature items, and 330 were grey. A multistage content analysis process was implemented to classify the rich literature body across multiple dimensions with further mapping, synthesis, and reporting.

Results: A taxonomy of pricing-related concepts was created. It classifies SaaS pricing aspects, affecting factors, and challenges facing SaaS providers. The findings and interpretations are summarized to emphasize the major research themes and practical challenges of SaaS pricing practices' transformation and provide further research guidelines in this area.

Conclusion: SaaS pricing is a maturing and prominent area of research that requires further investigation. The conducted MLR formed a clear picture of SaaS pricing research and practice and identified different SaaS pricing aspects and affecting factors. The study will enable both scholars and practitioners to assess the current state-of-the-art in research and practice.

1. Introduction

Software-as-a-service (SaaS) is a software licensing and distribution model in which software is hosted by service providers and is made available for customers over the Internet. According to the latest analytical reports, the SaaS market revenue is forecasted to grow to $104.7 billion in 2020 [1]. The real revenue could be even higher as the COVID-19 pandemic accelerates SaaS services' adoption by companies and individuals to address the increase in remote work and the demand for business agility [2]. By now, more than 82% of businesses use at least one SaaS solution, and the number of companies that use SaaS and the number of SaaS solutions used per company are both growing constantly. This rapid growth makes SaaS the largest public cloud computing market segment and the primary software licensing and delivery model globally [3].

A base case scenario of the SaaS model assumes remote access to the software on a subscription basis rather than buying a license and installing the software on local computers and servers. The software itself is owned, developed, and managed by an SaaS provider. Two types of SaaS providers can be identified: traditional software vendors and tech companies (e.g., SAP, Google, Adobe), and new "born-in-the-cloud" SaaS companies who usually have just one flagship SaaS solution, such

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as Salesforce add Dropbox [4]. The fast transition of large companies towards these models, and the explosive growth in the number of new pure SaaS companies, is driven by significant benefits offered for SaaS providers. Still, these benefits come with business and technical challenges. The market landscape changes, which requires software companies to design and implement fundamentally different business models, product management practices, and development processes for their SaaS solutions, from those at the age of pure on-premise software products [5].

Pricing is an essential, crucial, and challenging element of product management and business strategy. By SaaS pricing, we denote the overall scope of the decisions, practices, underlying conditions, and processes in determining the monetary value of the offered SaaS service. Even a small change in the SaaS service price may significantly, positively or negatively, impact companies' financial performance. Defining the price for a product or a service is part of a comprehensive pricing management strategy that companies have to handle. Efficient pricing management requires sophisticated decision-making and analytics, as well as coordination and finding compromises between the many business functions involved. Many companies find this challenging [6].

Facing the above-mentioned pricing challenges, large software and tech companies employ economists who cooperate with product and project managers to address all their products' pricing challenges, including SaaS solutions [7]. However, a wide range of newly established SaaS providers, most of which are small- and medium-sized enterprises, have the resources and knowledge to make informed decisions on pricing strategy, tactics, and implementation operations. Inconsistent knowledge of SaaS pricing and complications in establishing and managing all pricing-related processes and practices results in a scattered and under-managed pricing process for many SaaS providers [6].

Haphazard SaaS pricing in the software industry is mirrored in academic literature. Since its inception, SaaS and all its associated aspects, including pricing, have become topics of interest for scientists in various research domains. SaaS pricing has gained significant attention in many research domains, including economics, management science, and marketing, as well as software engineering and computer science. However, the lack of a single "home" for studies on software and SaaS pricing in the academic community has resulted in isolated pricing-related studies with diverse and inconsistent approaches and recommendations. As a result, the current theory does not sufficiently assist practitioners in selecting from among the many options while designing and implementing the pricing of their SaaS solutions [8]. The primary research goal was to address this issue by answering the following three research questions: (1) *What is the current status of academic research and practical expertise in SaaS pricing?* (2) *How is SaaS pricing defined and disseminated by scholars and practitioners?* and (3) *How the research outcome and practical expertise can support SaaS providers in pricing their products?*

We searched for articles from both pricing practitioners and scholars. Combining these two sources of publications in this systematic multivocal literature review, allowed us to explore existing pricing frameworks and systematize the diverse range of recommendations and guidance grounded in research or practical experience. Following the guidelines on conducting multivocal literature reviews (MLRs) [9], we explored the available academic literature on SaaS pricing across various digital libraries and databases, as well as accessing materials produced by practitioners and industry experts outside the traditional academic community. The body of literature included **57** items of white literature (WL) and **330** items of grey literature (GL) published since 2001, when the concept of SaaS was introduced[1].

SaaS pricing research should provide a body of knowledge that offers appropriate SaaS pricing solutions and designs to product managers based on the parameters and objectives of a given situation. Putting together pricing frameworks, step-by-step solutions, and easy-to-use decision support mechanisms, could assist product managers and, in the long run, improve industrial SaaS pricing practices. This study contributes to the development of a body of knowledge on SaaS pricing by comparing and systematically analyzing the existing literature. The study matches industrial state-of-the-practice with academic state-of-the-art to make suggestions about promising paths for future research. Practitioners can benefit from understanding how industrial practices can be improved from these academic studies. The study also provides a better understanding of the interplay between software product management and software engineering.

The rest of this article is organized as follows. Section 2 discusses related work. Section 3 illustrates the methodology, research questions, and scope of our systematic study. Section 4 presents the results of our study. Section 5 discusses the results and limitations of our study. Section 6 concludes this article by presenting the findings, implications, and directions for future work.

2. Background and rationale

In this background section, we define the concept of SaaS and identify the issues of SaaS pricing. We also give a brief overview of existing SaaS pricing reviews and overviews.

2.1. The concept of SaaS

The idea of centralizing computational power and application hosting with machine time-based paid access, was first implemented by computer service bureaus in the 1960s [10]. The appearance of personal computers enabled the replacement of this model by the on-premise software model, which offered the ability to purchase a perpetual license and install a copy of the software on a local computer. However, the development of the internet has led to another delivery model of commercial software – the software lease model. The consumer is entitled to use the software product via the Internet or a "thin client," where the software is owned and operated by an application service provider (ASP) [11]. The provider can produce the software itself or buy the software from a developer. The provider operates and maintains the servers that run the software. Before the year 2000, this was not very widely accepted or adopted by small and medium-sized enterprises (SMEs), individuals, or large corporations despite the attractiveness of the idea itself [12]. In the late 2000s, with further development of the Internet, the ASP model re-emerged under the notion "software as a service" and as part of the cloud computing paradigm.

The acronym SaaS was first invented in 2001 by the Software & Information Industry Association (SIIA) [13] to describe a model where: *"… the application, or service, is deployed from a centralized data center across a network – Internet, Intranet, LAN, or VPN – providing access and use on a recurring fee basis. Users "rent," "subscribe to, "are assigned," or "are granted access to" the applications from a central provider."* However, the most common and generally accepted definition is the one presented in 2011 by the United States National Institute of Standards and Technology (NIST) [14]. First, NIST defined cloud computing in general as: *"…a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."* Further, SaaS itself is defined as one of three models. The other two are platform-as-a-service (PaaS) and infrastructure-as-a-service (IaaS).

Specifically, SaaS is "the capability provided to the consumer to use the provider's applications running on the cloud infrastructure" [14]. The applications are accessible with various devices through a web browser, a thin client interface, or an application. The consumer does not manage or control the underlying cloud infrastructure, except for limited user-specific application configuration settings. NIST also lists

[1] Selected WL and GL items, including extracted fields, are presented in the online supplementary appendix available at https://bit.ly/3dQXK9M.

the following five essential characteristics of SaaS [14]: on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service. The NIST definition of SaaS is quite technical and focuses primarily on the deployment and delivery aspects of the SaaS model.

Outside the information systems and software engineering domains, scholars have discussed economics, business, and behavioral issues of using various forms of online software-intensive services from a non-technical perspective. Instead of SaaS, a variety of names are used in publications without providing a clear definition and relying on a commonsense understanding (i.e., "cloud services" [15], "information services" [16]).

2.2. SaaS pricing

Consolidating the variety of different definitions provided by scholars and practitioners [17–19], we define pricing as the process of decision-making in determining the monetary compensation and related conditions of the goods and services the customer is offered. Pricing serves as an essential bridge between different business functions (i.e., product management, revenue management, cost management, retention management) and business units (i.e., R&D, production, sales, marketing). The decision-making in pricing is based on an integrated analysis of different perspectives and streams of information. Pricing is an essential element of the business model and product strategy. All this applies also to SaaS. Before exploring existing knowledge on the role of pricing in SaaS companies, we overview the concept of pricing and its application to the software industry in general.

While the price has been the central element of economic theories for centuries, the concept of pricing as a managerial discipline and business function dates back to the late 1970s. It arose as a result of the deregulation of the airline industry which provided flexibility for airline companies in defining prices for airline tickets [20]. Back then, pricing was considered as a part of revenue (or yield) management, defined as the processes and practices of selling "the right inventory unit to the right type of customer, at the right time, and for the right price" [21]. Efficient revenue management required comprehensive decision-making regarding these four "right" aspects, with the goal of maximizing revenue streams. Since then, a considerable amount of management literature on revenue management has been published, exploring its evolution and variation among industries and even among companies within the same industry [22].

Another important pillar of any business model and strategy is cost management [23]. The cost structure and its management practices vary significantly across industries. Aligning revenue and cost management practices is essential to ensure profitability and long-term sustainable development.

The software industry has unique characteristics of revenue, pricing, and cost management. First, revenue management in the software industry is mostly about defining "right pricing" and "right customers". The timing of production and inventory are of little importance. Second, most software companies have a considerable disparity between fixed costs and variable costs, which creates supply side economies of scale [24,25]. These two unique characteristics make pricing a key driver for market success and revenue growth for software companies.

While the commercial success of software companies is very dependent on an adequate pricing strategy; decisions on designing and implementing a pricing strategy have always been challenging for software companies [26]. If there is a lack of focus in pricing at strategic, tactical, or operational levels, the product and the company are likely to fail. The transition towards the SaaS business model has enabled new opportunities for software companies in software development, delivery, and operation. These opportunities have implications for pricing by creating and magnifying the number of pricing design, experiment, and control methods available. These methods include, for example, recurring subscription fees, new methods to ensure efficient price discrimination, and real-time usage tracking [27]. However, these new opportunities can also cause obstacles for companies when old pricing principles and practices become obsolete, and companies' understanding of how the new ones should be designed is unclear [28].

The unique characteristics of software as a product and the software market [29], require tighter alignment between pricing management and development processes than in any other industry. Moreover, pricing is considered as an integral part of software product management and corresponding responsibilities [30,31].

2.3. SaaS pricing reviews

The crucial importance of pricing for businesses has promoted research in this area. The number of academic publications on SaaS pricing topics has grown as the adoption of this new model has increased. However, as it will be proved below, SaaS pricing research remains at a level far below the importance of this topic for practice. there is an evident lack of a systematic investigation of the development and current state of this research area. While background sections of several research papers (i.e., [32,33]), and several literature reviews (i.e., [34,35]) have highlighted certain aspects of SaaS pricing research, they have covered only a narrow part of the literature. In fact, we were able to identify only one systematic review that examined SaaS pricing [36]. However, this publication had a broader scope, explored all three cloud computing models, and was based on a market research perspective. In this study, we target the lack of systematic reviews on SaaS pricing. Our aim is to obtain a clear picture of SaaS pricing, its different approaches, and pricing methods employed.

3. Research methodology

This MLR focuses on SaaS and its pricing across various research domains and types of studies. Our objective is to identify the state-of-the-art and the state-of-practice in SaaS pricing. We also want to provide a basis for further research in SaaS pricing. We do not limit the scope of the study to a systematic review of academic publications (white literature, WL). Instead, we also incorporate an extensive body of grey literature (GL) in the analysis. Following [37,38], we refer to publicly available knowledge artifacts in both digital and printed formats, which can also be produced outside academic publication channels. GL publications considered for this research include, but are not limited to, discussion and white papers, blog posts, reports, web-pages, and magazine articles. WL includes publications in academic venues that are prepared through a formal peer-review process. These include scientific journal articles, conference proceedings, working papers series, and monographs.

The research protocol for the study is based on the guidelines for performing systematic and multivocal literature reviews and mapping studies [9,39,40]. The formal research process is presented in **Fig. 1**.

3.1. Research questions

To get an integrated, transparent, and fresh look into the research and practice of SaaS pricing, this study focused on answering three broad research questions *RQ1–RQ3,* with several clarifying sub-questions per question. Following the classification scheme discussed in [9,41]), all three research questions (with corresponding sub-questions) can be classified as exploratory ones.

RQ1: What is the current status of academic research and practical expertise in SaaS pricing?
RQ1.1: What is the total number of GL and WL publications on SaaS pricing and how has the number of publications evolved over time?
RQ1.2: Who are the leading scholars and practitioners in this area, and what venues are used for publications?

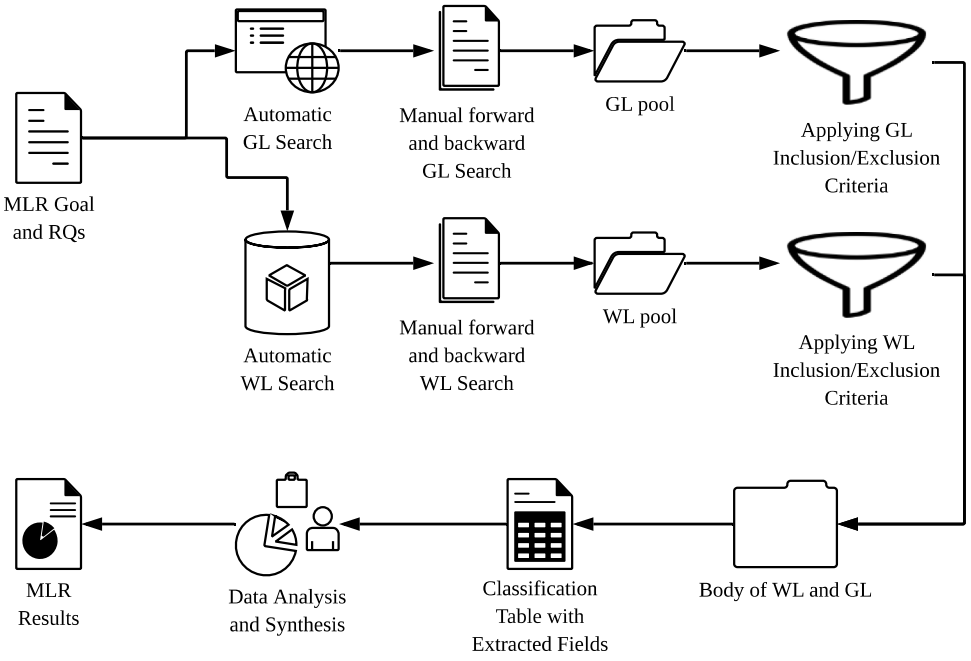


Fig. 1. MLR research process.

RQ1.3: What research and analytical methods have been adopted in the existing literature?
RQ1.4: What types of SaaS and SaaS markets are considered, and what companies are used as case studies?
RQ2: How has SaaS pricing been defined and structured by scholars and practitioners?
RQ2.1: What are the objectives and the role of pricing in the business of SaaS companies?
RQ2.2: What aspects of SaaS pricing are identified and addressed in the existing literature and how can they be classified?
RQ2.3: How is SaaS pricing structured and dissected in existing literature?
RQ2.4: How are SaaS pricing-related terms introduced and defined by scholars and practitioners?
RQ3: How can the research outcome and practical expertise support SaaS providers in pricing their products?
RQ3.1: What are the primary challenges facing SaaS companies in designing and implementing pricing?
RQ3.2: What factors affecting SaaS pricing are identified in the existing literature and how can they be classified?
RQ3.3: What SaaS pricing decision-support frameworks, if any, have been proposed and how do they vary?

The main idea behind ***RQ1*** is to provide a broad overview of SaaS pricing as an area of research and practice. The first two sub-questions (*RQ1.1* and *RQ1.2*) focus on evaluating the distributions of publications based on source type, date, authorship, and venues, as primary indicators of interest of scholars and practitioners and the area's level of maturity, narrative, and focus. From different perspectives, the next two sub-questions (*RQ1.3* and *RQ1.4*) assess how comprehensive and multifaceted the current knowledge and expertise in the area is.

With ***RQ2***, we identify how scholars and practitioners understand and systemize SaaS pricing. We start with identifying SaaS pricing objectives (*RQ2.1*) and aspects addressed in these publications (*RQ2.2*). We then assess how SaaS pricing can be structured (*RQ2.3*), and assess the common vocabulary of terms associated with SaaS pricing (*RQ2.4*).

The last research question ***RQ3*** investigates the practical implication of current knowledge on SaaS pricing and whether existing publications offer reliable support for decision-making. Firstly, we identify the main challenges and pain points of SaaS pricing (*RQ3.1*). Secondly, we identify factors that should be taken into account while defining and implementing the pricing of SaaS solutions (*RQ3.2*). Finally, *RQ3.3* identifies and compares existing decision-making frameworks.

3.2. Research scope

The overall primary research scope driven by the formulated research questions was identified using Population, Intervention, Comparison, Outcomes (PICO) criteria [40]:

Population: The notion of SaaS was first introduced in 2001[9]. Therefore, we looked for papers published between 2000 and Q3 2020. We did not limit the publication venue and considered both WL, that is, articles in peer-reviewed journals and conference proceedings, and GL, that is, textbooks, monographs, technical reports, blog posts, and discussion boards [9].

Intervention: We took the following actions to produce a relevant MLR: collected the prime body of literature, performed data extraction and classification, and conducted synthesis and analysis in a form that allowed us to provide answers to research questions *RQ1–RQ3*.

Comparison: The MLR study compared various issues and aspects regarding SaaS pricing across multiple dimensions. By analyzing the extracted information, we were able to summarize current knowledge in the field of SaaS pricing. We considered a cross-domain analysis of trends in research and practice, contribution, and challenges, as essential to our objectives. Both similarities and differences were used to identify the potential for further studies as well as to highlight improvements to SaaS pricing practices in the industry.

Outcome: The collected body of literature represented a wide-ranging coverage of existing studies and practical observations and ensured the comprehensiveness and authenticity of the study. The MLR provided a clear overview of SaaS pricing, including identifying the key challenges across domains and proposing further research directions to address them.

3.3. Source selection and search strategy

Defining the research questions and research scope was followed by determining sources and search strategy. Given the vast body of

literature regarding the research topic, the data collection routine was based on automatic queries across multiple scientific databases and digital libraries, as well as Google and Google Scholar search engines. The literature review consisted of two stages of searching. Search procedures were defined according to the PICO criteria. In the first stage, we collected WL using multiple scientific databases and digital libraries. In the second stage, we collected GL following a similar protocol, using the search engines mentioned above.

At both stages, we employed similar search query strings. They were constructed in order to retrieve the most relevant results on issues related to SaaS pricing. We defined the following search string as a combination of keywords and operators, and used it to perform searches in databases, digital libraries, and search engines:

("SaaS" OR "software-as-a-service") AND ("pricing" OR "price" OR "revenue" OR "costs" OR "fee" OR "charge").

The stopping criteria [9] were quite similar for both types of literature. We bounded the efforts by considering only the first 300 search database engine hits. Additionally, we ensured that, by that threshold, we had an evident theoretical saturation, and that relevant publications stopped appearing.

3.3.1. The first stage: WL search

We selected the following scientific databases and libraries that cover the most significant journals and conference proceedings: ScienceDirect, SpringerLink, Scopus, JSTOR, IEEE Xplore, and ACM Digital Library. The search procedure was conducted in September 2020. To ensure the exhaustiveness of the collected body of literature, we complemented the automated search with a backward and forward chaining manual search, using both the Google Scholar search engine. While we screened all identified initial studies for ACM Digital Library, IEEE Xplore, ScienceDirect, and JSTOR, we relied on the ranking algorithm for SpringerLink and considered only the first 300 items. However, relevant results stopped showing up after we screened the first 200 items, and we did not find any relevant items among the last 50 considered items. Scopus and Google Scholar were used for validation of the relevance of the collected body of literature and for capturing possibly missing items for the body of literature. For Google Scholar, we used the same principle as with SpringerLink and considered only the first 300 items. As in SpringerLink, we did not find relevant items among the last 50 items, which suggested that there was no need for a further search. All identified papers were first screened using the inclusion criteria (IC), which helped to identify papers that met the research scope:

1 Full texts of the paper are available,
2 Published in peer-reviewed journals or conference proceedings,
3 The study is not a duplicate of another study, and
4 The study covers any aspect of SaaS pricing.

To ensure that we were not missing any significant part of the study, we supplemented the automatic search across scientific databases and digital libraries with a so-called backward and forward chaining search for the first 50 publications, based on the number of citations in Google Scholar. This approach allowed us to identify papers that either did not focus on SaaS pricing, but still covered certain aspects and provided valuable insights, or used various synonyms of the term SaaS (i.e., "cloud services," "online services," or "information services") while discussing issues relevant to SaaS pricing. After the initial body of WL collection was completed, the following exclusion criteria (EC) allowed us to exclude irrelevant papers based on the full-text analysis:

1 The study does not specify in which way the research contributes to a better understanding of SaaS pricing or improving SaaS pricing practices,
2 The study does not provide clear evidence for its results, and
3 The study only reviews existing studies without providing substantial analysis and synthesis.

As a result, the final collected body of WL consisted of **57** items stored for further data extraction and analysis.

3.3.2. The second stage: GL search

Using the Google search engine instead of scientific databases and digital libraries, we employed the defined search query and explored the first 300 web resources identified. We manually evaluated them using the following IC:

1 The publication is publicly available on the Internet,
2 The publication is a standalone publication written under a real name or the name of an organization,
3 The publication content is original, and its length exceeds 250 words, and
4 The publication covers any aspect of SaaS pricing, apart from just the importance of pricing.

For those publications that met the ICs, we conducted a manual search using internal navigation and search mechanisms on the web site. Often, a web site contained more than one publication related to SaaS pricing. In the same way, we did not include short publications (usually blog posts in the form of a comment), nor did we include books, reports, or magazines as a single item of GL. We divided lengthy publications (i.e., books or magazine issues) into sections, chapters, or articles, in order to have a more homogeneous body of literature. Finally, we manually added 13 items of GL (white papers/reports) identified through the Google Scholar search that were not included in the cohort of WL. After the initial body of the GL collection was completed, we applied the following EC for a full-text analysis to exclude publications from those that have already been included:

1 The publication does not offer details on the design or implementation of SaaS pricing,
2 The publication does not include industrial cases or other factual evidence, and
3 The publication only reviews existing materials on SaaS pricing issues without providing conclusions with substantive value.

As a result, the final collected body of GL consisted of **387** items stored for further data extraction and analysis.

3.4. Data extraction and analysis

After we completed the search procedures, the body of literature consisted of **387** items. We employed a multistage procedure to extract the information needed to provide answers to the defined *RQs*. The list of extracted fields assigned with the *RQs* is presented in **As previously** mentioned, all three RQs along with corresponding sub-questions, are exploratory ones. Answers to the research sub-questions RQ1.1-RQ1.4 were obtained using statistical analysis. We calculated the required indicators, visualized them, and explored the received distributions and trends. Based on the analysis, we provided observations that were further used to define suggestions for further research and to answer RQ1 in general.

Providing answers to research sub-questions RQ2.1, RQ2.2, RQ3.1, and RQ3.2, required classification and synthesis in addition to the statistical analysis. The first part of the analysis was carried out as part of the data collection. Based on the extracted raw data, we constructed a classification for various SaaS pricing issues with a repeated extraction procedure to fill the classification tables. The second part of the analysis involved assessing the frequency distribution of various SaaS pricing issues and comparing them across different dimensions, including types

of publication and publication venue. Similarly to the first group of research sub-questions, we also provided observations that were further used to define research gaps and to propose research directions to fill the gaps.

Answers to the rest of the research sub-questions were mostly formed by combining, integrating, and comparing information extracted from the body of literature. Definitions for various terms associated with SaaS pricing were constructed (RQ2.4), and existing SaaS pricing taxonomies (RQ2.3) and decision-making frameworks (RQ3.3) were compared.

Table 2. If possible, data extraction was performed automatically; however, most of the of the fields required manual extraction. To mitigate subjectivity and biases, fields WL11/GL11–WL18/GL18 were extracted using a multi-step procedure as follows:

1 First, we manually filled in these fields with all the relevant information available in publications;
2 Second, we analyzed all collected manual inputs and systematized them. For most of these fields, we developed a classification. We were not able to do it for all fields as these issues were not discussed in the vast majority of publications and the collected info did not provide sufficient grounds for such generalization (i.e., SaaS pricing business role and objective, WL11/GL11); and
3 Finally, we read the publications for a second time to fulfill classification for the fields where we were able to develop classification and to ensure that we are not missing any relevant information.

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Table 1
Selected search sources and collected items.

Digital library/ database	Web address	Initial number of publications	Final number of publications	
			WL	GL
ACM Digital Library	http://dl.acm.org	12	7	-
IEEE Xplore	http://ieeexplore.ieee.org	107	15	-
ScienceDirect	http://www.sciencedirect.com	31	22	-
JSTOR	https://www.jstor.org	351	13	-
SpringerLink	https://link.springer.com	4 607	34	-
Scopus	https://www.scopus.com	278	37	-
Google Scholar	https://scholar.google.com	94 700	42	42
Google	https://google.com	121 000 000	-	302
Forward and backward search			10	21
TOTAL (without duplicates)			57	330

Table 2
Data extraction fields.

Research question	Extracted field code	Extracted field
RQ 1.1	WL1 / GL1	Publication title
	WL2 / GL2	Publication year
RQ 1.2	WL3 / GL3	Publication author(s)
	WL4 / GL4	Publication venue / source name
	WL5 / GL5	Publication type
	GL6	Source type
RQ 1.3	WL7 / GL7	Research / analytical approach
RQ 1.4	WL8 / GL8	SaaS category
	WL9 / GL9	Types of customers
	WL10 / GL10	SaaS companies
RQ 2.1	WL11 / GL11	SaaS pricing business role and objectives
RQ 2.2	WL12 / GL12	Considered SaaS pricing aspects
RQ 2.3	WL13 / GL13	Proposed SaaS pricing schemes and typologies
RQ 2.4	WL14 / GL14	SaaS pricing-related terms and concepts, and their definitions
RQ 3.1	WL15 / GL15	SaaS pricing challenges
RQ 3.2	WL16 / GL17	Factors affecting SaaS pricing
RQ 3.3	WL17 / GL17	SaaS pricing decision-making frameworks structure

used to define research gaps and to propose research directions to fill the gaps.

Answers to the rest of the research sub-questions were mostly formed by combining, integrating, and comparing information extracted from the body of literature. Definitions for various terms associated with SaaS pricing were constructed (RQ2.4), and existing SaaS pricing taxonomies (RQ2.3) and decision-making frameworks (RQ3.3) were compared.

4. Analysis and results

In this section, we analyze the collected body of WL and GL from different perspectives, defined by the research questions specified in **3.1**. Results are summarized in a form of 14 Preliminary Empirical Conclusions (PECs).

4.1. Current status of academic research and practical expertise on SaaS pricing

RQ 1.1. What is the total number of GL and WL publications on SaaS pricing, and how has the number of publications evolved over time?

The distribution of the studies by year, and belonging to WL or GL is presented in **Fig. 2**. It shows that SaaS pricing is a widely discussed topic among scholars and practitioners. In 2003, the first academic paper devoted to the issue of pricing strategy for SaaS was published. The authors used the term "web-enabled application services" [42]. However, the second paper on SaaS pricing, in chronological order, was only published three years later, in 2006. Starting that year, SaaS became the primary term for the new and promising licensing and delivery model. The first GL publication we found dates back to 2009. However, it is possible that some publications have been removed from websites or were published on websites that no longer exist.

The number of publications, including both GL and WL, has been growing annually (for 2020 we identified only publications published between Jan-Sep). However, the trend in the number of publications is opposite for GL and WL. While the number of WL studies is declining, the number of GL studies is increasing. The total number of GL publications far exceeds the number for WL. Currently, most publications (**330** items, 85%) on SaaS pricing belong to GL. Even though we excluded publications with fewer than 250 words, most GL publications are still relatively short blog posts expressing the author's opinion on a particular aspect of SaaS pricing.

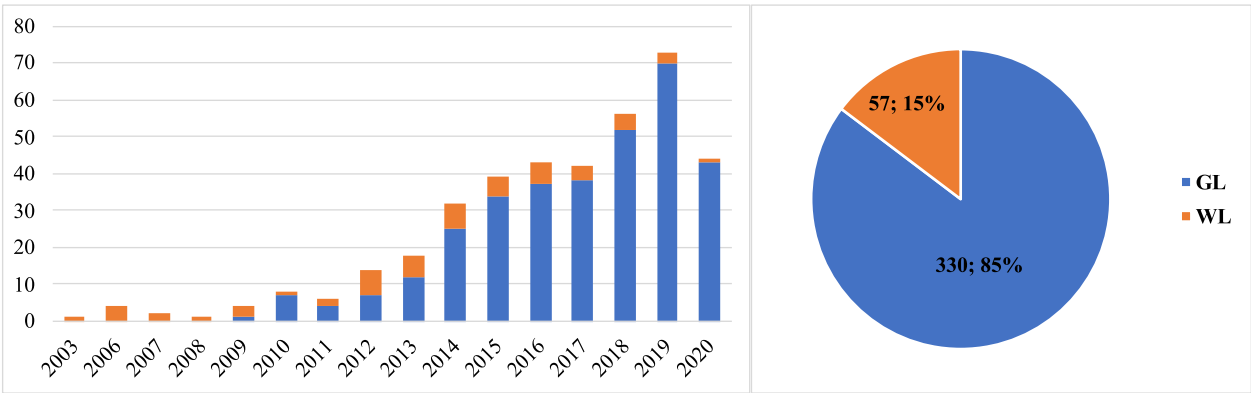


Fig. 2. Distribution of the collected publications by year (left) and publication type (right).

PEC #1: The number of WL publications per year is declining, indicating a decline of interest by scholars to pursue this topic. However, the number of GL publications is growing.
PEC #2: There is an imbalance between the amount of GL and WL, with a low number of publications by scholars and a high number of publications by practitioners.

RQ 1.2. Who are the leading scholars and practitioners in this area, and what venues are used for publications?

Table 3 lists leading scholars and practitioners in the field of SaaS pricing. Lists of scholars and practitioners by themselves do not provide much insight besides naming who is working in the field. It is noteworthy that only one scholar was a co-author in more than five publications. Among practitioners, we noted a group of experts actively publishing and sharing their experiences on various SaaS pricing-related topics. The analysis of the awareness of the scholars with regard to the practitioners' activities and vice versa, brought insightful results. We did not find any citations of GL publications in academic papers. Similarly, we found no citations of WL publications in GL publications. Practitioners frequently mentioned different studies (i.e., behavioral economics [43–45], and general marketing [44,46]), but never dedicated studies on SaaS pricing. Additionally, we did not find any GL publications co-authored by scholars.

Table 4 lists the leading publishing venues for SaaS pricing research and Internet resources used by practitioners. The rest of the WL venues not listed in the table have just one publication; for GL, we listed all venues with the number of publications above five. The list of venues

Table 3
List of leading scholars and practitioners.

WL Author name	Number of publications	GL Author name	Number of publications
Ojala, A.	5	Campbell, P.	30
Li, M.	4	Murphy, L.	17
Zhang, Z.	4	Poyar, K.	15
Laatikainen, G.	3	Tunguz, T.	12
Ma, D.	3	Lemkin, J.	10
Nan, G.	3	Forth, S.	9
Buxmann, P.	2	Mele, C.	9
Cao, R.	2	Shelley, E.	6
Cheng, H.	2	Pena, N.	6
Jiang, Z.	2	Smith, E.	6
Lehmann, S.	2	Balaji, S.	6
Mehra, A.	2	Harvey, K.	6
Bala, R.	2	Wise, W.	5
Seidmann, A.	2	Dimova, P.	5
Sun, W.	2	Guo, V.	5
Tan, Y.	2		
Zheng, Y.	2		

confirmed the interdisciplinary nature of the research in SaaS pricing. While we found that examples of multidisciplinary conferences and journals that accept publications from different research domains, research on SaaS pricing, in general, was fragmented, and there were very few cross-domain citations. Our findings also showed that studies on SaaS pricing can be published in highly ranked journals in management science and information systems. At the same time, the high diversity of publication venues of existing SaaS pricing studies and the absence of special issues in journals, may indicate a lack of venues for SaaS pricing research.

We also classified both WL and GL publications by type (Fig. 3). We identified three types of WL: journal article, conference proceedings, and working paper. The variations of GL appeared to be more numerous. We classified all GL publications into the following four groups: Webpage/Blog post, Presentation/Infographics, White paper/Company report, and Book section/Magazine article. Some companies position their white papers and reports as books or magazines of their own (i.e. [47,48]). Despite that we classify them as White paper/Company report. A Book section/Magazine article is published by an independent publisher in our classification. In WL, the share of conference proceedings (25 items, 43%) slightly exceeded the number of journal articles (27 items, 47%) and working papers (6 items, 10%). The highest share of GL was found in blog posts (307 items, 92%).

We also classified GL based on the source type. We identified the following six sources of SaaS pricing publications: SaaS provider blog, consulting company blog/journal, venture capital company blog/journal, media website/industrial magazine, industrial conference/event presentation/speech, and personal blog. The distribution of GL publications based on the source type is presented in Fig. 5. Consulting companies' blogs and reports had the most publications (157 items, 47%). While leading consulting companies (i.e., McKinsey, Bain, PWC) published in this area, consulting companies actively sharing knowledge in pricing are relatively small and are usually highly specialized. The second largest share of GL publications (86 items, 26%) was prepared mostly by SaaS companies offering help for SaaS providers and other digital companies, in pricing- and revenue-related activities (i.e., billing, customer analytics, pricing design)

PEC #3: Absence of any cross-references between WL and GL and scarcity in cross-references between research domains.

RQ 1.3. What research and analytical methods have been adopted in the existing literature?

Both WL and GL publications employ a diverse range of research and analytical approaches. We used the following classifications for our research approaches (**Fig. 6**):

1 Case study and in-depth survey,
2 Design science and action research,

Table 4
List of leading publication venues and sources.

Venue / source name	Venue / source type	# of publications
WL		
Social Science Research Network (SSRN)	Working Paper Series	5
International Conference on Software Business	Conference	4
Information Systems Research	Journal	2
International Conference on Information Systems	Conference	2
International Journal of Advanced Computer Science and Applications	Journal	2
Journal of Management Information Systems	Journal	2
Journal of Revenue and Pricing Management	Journal	2
GL		
OpenView Partners	Venture capital company	37
PriceIntelligently	Consulting company	36
Sixteen Ventures	Consulting company	17
Chargify	SaaS provider	14
PayMotion	SaaS provider	14
Redpoint Ventures	Personal blog	12
Fusebill	SaaS provider	12
SaaStr	Consulting company	11
Chargebee	SaaS provider	11
Cobloom	Consulting company	10
Klood Digital	Consulting company	10
Software Pricing Partners	Consulting company	9
Medium	Media website / Industrial magazine	9
ChartMogul	SaaS provider	7
Lighter Capital	Venture capital company	6

3 Quantitative research,
4 Simulation and stochastic modeling, and
5 Overview, literature reviews, and mapping studies.

In line with this, the largest share of WL publications (31 items, 54%) use economic models to investigate what pricing strategies that software companies who offer SaaS should follow, using market determinants, product characteristics, and market equilibrium as factors. The second-largest share (11 items, 19%) of academic publications included literature reviews on SaaS pricing or overviews of industrial practices. Most of these studies provided taxonomies of SaaS pricing practices with a discussion and basic descriptive statistical analysis. All studies except one [36] were conducted in a non-systematic way.

We used the following typology of analytical approaches based on the analysis of the GL pool:

1 Experience sharing – the prime goal of the publication is to clarify certain SaaS aspects or challenges related to SaaS pricing based on the authors' personal experience or inside information from a particular SaaS company.
2 Opinion sharing – the prime goal of the publication is to clarify certain SaaS aspects or challenges related to SaaS pricing based on rational reasoning or outside observation of particular SaaS companies.
3 Overview and systematic observation – the prime goal of the publication is to list, overview, or compare various pricing aspects and methods.
4 Solution proposal – the prime goal of the publication is to provide a systematic recommendation or framework to support the SaaS pricing decision-making processes.
5 Survey and questionnaire – the prime goal of the publication is to share the results of a survey with SaaS companies.

The distribution of GL publications using the analytical method is presented in Fig. 7. Only 21 (6%) GL publications framed their ideas, knowledge, and experience in any structured way that could be called a solution (framework or decision-support tool). Many publications were generated by consulting companies and individual experts. While they were ready to share for free their observations and provide advice, systematic solutions developed by them were available from a paid consulting service or paid training. Additionally, just 6 (2%) publications presented the results of the performed surveys with other experienced professionals and SaaS executives.

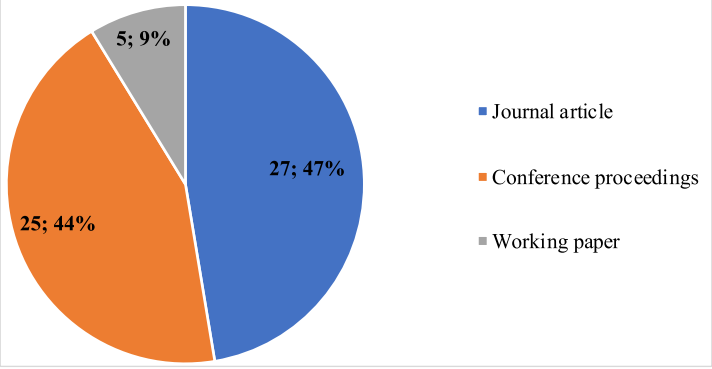


Fig. 3. Distribution of the publication by type for WL.

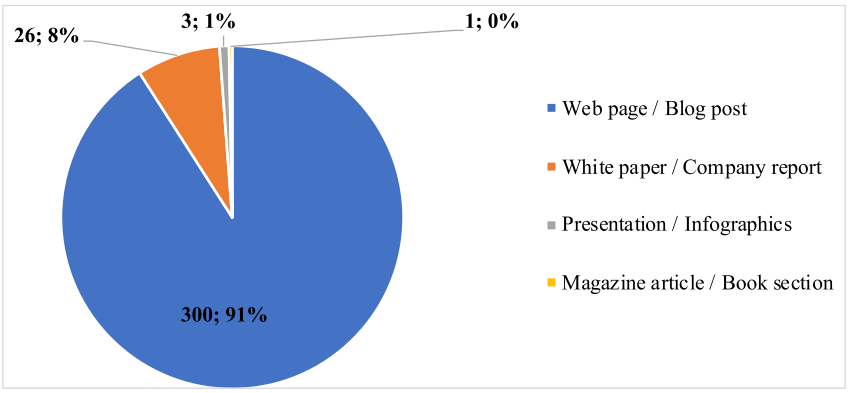


Fig. 4. Distribution of the publication by type for GL.

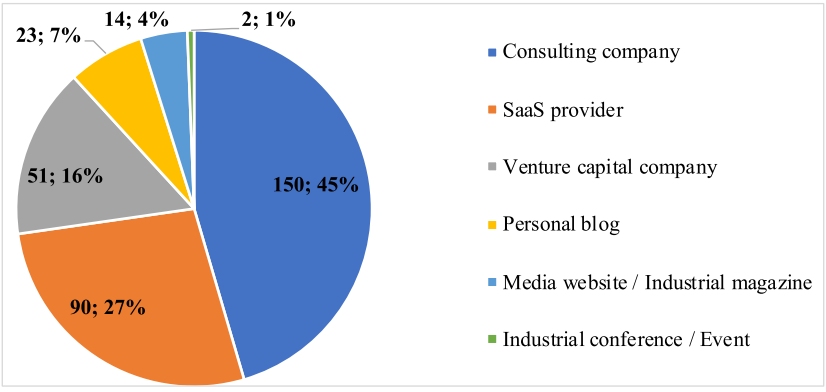


Fig. 5. Distribution of the GL publication by source type.

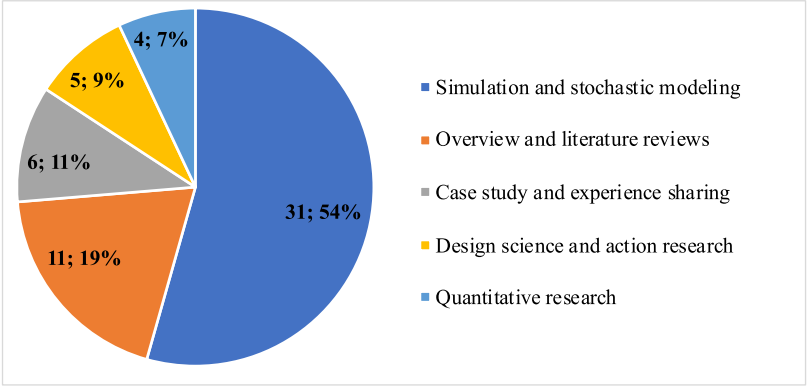


Fig. 6. Distribution of the WL publication by research approach employed.

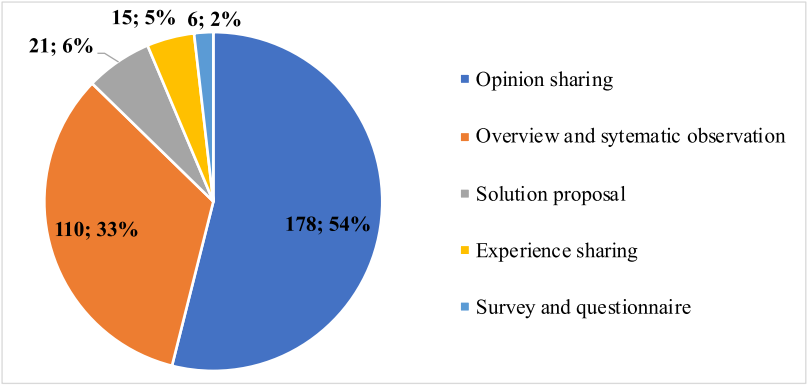


Fig. 7. Distribution of the GL publication by analytical approach employed.



Fig. 8. A word cloud for the companies used as examples in GL publications.

PEC #4: The research methods employed in in WL are mostly limited to theoretical economic modeling and non-systematic reviews.

PEC #5: Both GL and WL lack quantitative studies

RQ 1.4. What types of SaaS and SaaS markets are considered, and what companies are used for case studies?

With few exceptions (i.e., [28,49]), scholars did not specify the market and SaaS type that their research focused on. Additionally, scholars rarely provided illustrative examples of real-world SaaS companies, or they did it anonymously. Examples of SaaS companies that were briefly mentioned, were mostly limited to market leaders, such as Salesforce, Google Apps, or Oracle. Scholars often seemed to lack a realistic picture of the SaaS market, which is populated by a diverse range of small- and medium-sized SaaS companies in addition to the software giants.

However, practitioners frequently grounded their observations and classifications on the pricing of real SaaS companies and sometimes assessed the evolution of pricing in these companies (i.e., Intercom in [50]). They also often specified the market and product type of the SaaS solution being discussed and reflected on how these characteristics affect pricing. The range of SaaS companies of different types mentioned in GL is diverse. In total, we identified references to 151 SaaS companies. Visualization of their mentioning frequency is shown in **Fig. 8**.

PEC #6: Unlike practitioners, scholars tend to avoid contextualization of the research, assuming homogeneity of SaaS solutions, consumers, and markets.

4.2. Definition and structure of SaaS pricing by scholars and practitioners

RQ 2.1. What are the objectives and the role of pricing in the business of SaaS companies?

Existing studies on general pricing management revealed the vast range of objectives that can be targeted by product and service companies in different industries and markets [51,52]. We adopted their classification of pricing objectives in the SaaS context. They included 13 objectives across three broad categories: *Market, Company*, and *Customer*. We further assessed which of these objectives were identified as relevant within the collected body of literature. Table 5 illustrates these findings, and heat mapping, based on frequencies, allows visual assessment of their relevance.

We were not able to find any WL publication dedicated to an in-depth analysis of possible SaaS pricing objectives. Most academic studies briefly mentioned pricing objectives but often limited them to ensuring profitability and increasing market share (i.e., [53,54]). Practitioners provided a broader, more comprehensive view of pricing objectives (i.e., [47,55]). However, these views were often based only on personal opinions and non-systematic observations. Therefore, scientific research is needed in this area to systematize the diverse range of opinions and observations.

The role and organization of pricing in SaaS companies have not been explicitly addressed in academic literature and are seldom discussed in publications by practitioners either. Yet, we found GL publications (i.e.,[48]) that addressed this topic. Another discussion on the role of pricing in the organizational context proposed in [6], suggested that pricing is a bridge between a different business function and

Table 5
SaaS pricing objectives.

		WL Items	WL %	GL Items	GL %
Market	Market share maximization	13	23%	17	5%
	Welfare maximization	4	7%	0	0%
	Creating entering barriers	1	2%	0	0%
	Winning the competition	6	11%	18	5%
Company	Revenue/profit maximization	42	74%	92	28%
	Cost coverage	8	14%	46	14%
	Sustainable development	1	2%	26	8%
	Capacity utilization	0	0%	1	0%
Customer	New customer acquisition	0	0%	27	8%
	Customer churn reduction	1	2%	23	7%
	Monetization of existing customers	0	0%	27	8%
	Fair matching customers value perceived	7	12%	33	10%
	Piracy reduction	3	5%	1	0%

recommends organizing a special pricing committee.

PEC #7: Existing academic studies provide a quite narrow view of the objective of SaaS pricing. They do not describe the role of pricing in the organizational context. Practitioners deliver a more comprehensive, but still disorganized vision of pricing roles and objectives.

RQ 2.2. What aspects of SaaS pricing are identified and addressed in the existing literature?

The identified GL and WL cover a wide range of SaaS pricing aspects. We categorized all these pricing aspects into four categories. The first three categories are related to organizational levels and are titles as *Strategic Level, Tactical Level and Operational Level.* The fourth category of *Overall pricing* cover publications that discussed more general issues, including pricing concepts, trends, and ethical issues. Within these four categories, we identified 12 pricing aspects. The classification of SaaS pricing aspects is presented in Table 6.

Scholars in WL and practitioners in GL mainly focused on aspects that we assigned to Strategic level and accorded other categories less attention. WL covered almost no aspects related to pricing tactics and pricing processes. A diverse range of recommendations on Tactical level aspects grounded in psychology, sociology, and behavioral science are widely discussed in GL. However, scholars did not approach these with verification and systematization. Similarly, pricing strategy design, implementation, and control have not been popular topics in SaaS pricing research.

PEC #8: Research on SaaS pricing has mainly focused on general pricing issues and strategy design. There is little research on pricing tactics and organizational processes in SaaS pricing as well as on the evolution and trends in SaaS pricing.

RQ 2.3. How is SaaS pricing structured and dissected in existing literature?

We identified nine different approaches to structure SaaS pricing. Four of them were proposed by practitioners (S1-S4) and three by scholars (S5-S9). These structural frameworks are presented in Table 7. The perspectives of the identified approaches vary. The first two (S1, S2) propose a typology of pricing methods as a pricing configurator. The authors identify the main SaaS pricing dimensions (parameters) and identify the variety of pricing methods within each of these dimensions. As a result, SaaS pricing is considered as a portfolio of selected pricing mechanisms. Frameworks S4–S9 classify pricing from an organizational point of view and define various areas. Most of them provide examples and typologies for identified pricing components. To assess the complexity of the proposed approaches, we explored what SaaS pricing aspects they cover (Table 8).

PEC #9: Scholars and practitioners introduced several approaches aimed to structure and dissect SaaS pricing from several perspectives. However, the proposed approaches are inconsistent with each other, and no information on their actual usage by SaaS providers has been disclosed.

RQ 2.4. What SaaS pricing-related terms are introduced and how are they defined by scholars and practitioners?

Our review revealed a confusing state of terminology in SaaS pricing research and practice. We were able to identify more than 10 different terms or concepts associated with SaaS pricing, including pricing

Table 6
SaaS pricing aspects.

		WL Items	WL %	GL Items	GL %
Overall pricing	Theory, concepts and trends	34	60%	13	4%
	Monetary and business goals and objectives	1	2%	54	16%
	Ethics and compliance	4	7%	2	1%
Strategic level	Competitive research and market positioning	11	19%	21	6%
	Market segmentation and value proposition	6	11%	71	22%
	Pricing structure, strategies and models	53	93%	157	48%
Tactical level	Offering design, versioning and bundling	6	11%	207	63%
	Transparency, promotion and communication	2	4%	95	29%
	Customer acquisition, retention and usage analytics	2	4%	19	6%
Operational level	Ownership, control and decision-making	4	7%	39	12%
	Performance measurement, testing and evolution	3	5%	71	22%
	Resources and costs planning and management	9	16%	32	10%

Table 7
Composition of SaaS pricing structure approaches

Code	Composition	Ref.
S1	Six pricing parameters: 1) Formation of price: price determination, degree of interaction 2) Structure of payment flow: single payment, recurring payments, combinations 3) Assessment base: number of pricing components, usage-dependent, usage-independent 4) Price discrimination: 1st degree, 2nd degree, 3rd degree, multi-dimensional 5) Price bundling: offer, product, degree of integration, price level 6) Dynamic pricing strategies: penetration pricing, follow-the-free strategy, skimming strategy	[56]
S2	Seven pricing dimensions: 1) Scope: granularity of the offer (package vs. unbundling) 2) Base: information base that dominates the pricing decisions (performance-based, cost-based, value-based, competitor-based) 3) Influence: ability of buyers and sellers to influence the price 4) Formula: the connection between price and volume 5) Temporal rights: the length of the time period when the user can use the offering 6) Degree of discrimination: offering product/service for different buyers for different prices 7) Dynamic pricing strategy: changing the prices over time based on various factors	[57]
S3	Two-staged framework: 1) Pre-purchase phase: communication and transparency (competitors, vendor, market info); 2) Post-purchase phase: dynamism and service (loyalty loop, value proposition, pricing reassessment)	[49]
S4	Five pricing layers: 1) Value creation (efficiency, novelty, complementarily, lock-in) and Business case (ROI, cost price, price margin, KPI pricing), 2) Pricing structure (pricing model, software, services, billing & metering system, SLA), 3) Price and value communication (marketing channels, price list, feature list, service list value, proposition), 4) Price policy and Sales mechanism 5) Price level	[59]
S5	Four pricing segments: 1) Pricing strategy: pricing goals, portfolio analysis, overall strategy, segment-specific strategy, 2) Price formulation: overall pricing and policies, segment-specific pricing and policies, pricing data and rules, 3) Transaction management: opportunities and requests, prioritization and allocation, quoting and deal management, policy enforcement, 4) Performance management: compliance, pricing performance measurement and management, sales force and partner enablement	[58]
S6	Six-segment pricing canvas: 1) Customer segments (personas, needs) 2) Value proposition (the basis for pricing, price objects) 3) Cost structure 4) Competitors and market 5) Pricing strategy (pricing goals, positioning, differentiation, bundling, psychological price levers) 6) Price model (price objects, metrics, price points, price levers, price model, price validation)	[60]
S7	Three-component: 1) Pricing Models: the method by which a user pays to use your product and for how much (7 types) 2) Activation Models: the method by which a user starts using your product (5 types) 3) Pricing Strategies: the way your pricing model is presented and marketed (8 types)	[45]
S8	Four components: 1) Strategy: what is the goal of the price? 2) Philosophy: how does the company price relative to costs? 3) Structure: what is the pricing rubric? 4) Positioning: how best to communicate the price?	[61]
S9	Two-component pricing: 1) Pricing model (10 types) 2) Pricing strategy (5 types)	[62]

strategy, pricing model, pricing structure, pricing policy, pricing approach, pricing scheme, pricing philosophy, pricing mode, pricing practices, and pricing pyramid. The most widely used terms are *pricing strategy* (35% of GL, 57% of WL), p*ricing model* (36% of GL, 34% of WL), and p*ricing structure* (16% of GL, 9% of WL). Both scholars and practitioners often juggled with these terms and individually used their own terms in their publications without providing clear definitions. Table 9 summarizes the definitions and typologies we found.

Many publications in both WL and GL pools, provided lists of possible pricing strategies and methods, with hints about their applicability in different contexts (i.e., [63]). Such WL publications often lacked illustrative examples and discussion on the applicability of the identified or proposed pricing strategies/tactics in real-world environments. In comparison, GL publications tended to be less systematic and often did not provide clear definitions for discussed methods; still, they offered a broader range of such methods and almost always discussed real companies that had implemented such methods. Both WL and GL publications lacked systematic investigations of the applicability and incompatibility of these methods, depending on different market structures, product and company characteristics, and customers.

PEC #10: Scholars and practitioners seem to lack a coherent vocabulary of terms associated with SaaS pricing.

4.3. Practical implication of existing publications

RQ 3.1: What are the prime challenges facing SaaS companies in designing and implementing pricing?

Both scholars and practitioners repeatedly stated that pricing, especially in the case of digital goods, including SaaS, is a complex area. The majority of SaaS pricing studies, including theoretical reviews, quantitative and case studies, as well as microeconomic ones, did not clearly specify the challenges they aimed to address. In total, we were able to extract just five challenges briefly mentioned in GL and WL, as shown in Table 10.

PEC #11: Little is known about the challenges facing SaaS companies while designing and implementing SaaS pricing.

RQ 3.2: What factors affecting SaaS pricing are identified in the existing literature?

Table 8
SaaS pricing aspects coverage by structure approaches

		S1	S2	S3	S4	S5	S6	S7	S8	S9
Overall pricing	Theory, concepts and trends									
	Monetary and business goals and objectives				●		●			
	Ethics and compliance									
Strategic level	Competitive research and market positioning				●	●	●			
	Market segmentation and value proposition			●	●	●	●			
	Pricing structure, strategies and models	●	●	●	●	●	●	●	●	●
Tactical level	Offering design, versioning and bundling	●	●	●	●	●	●	●	●	
	Transparency, promotion and communication			●	●					
	Customer acquisition, retention and usage analytics									
Operational level	Ownership, control and decision-making									
	Performance measurement, testing and evolution			●						
	Resources and costs planning and management									

Table 9
Definitions of SaaS pricing-related terms.

Term	Definition	Typology	Ref.
Pricing strategy	"Pricing strategy is the way you price your products based on various factors, such as costs, business goals, market segment, the ability of consumers to pay, and the value you deliver" "A strategy is a plan you use to achieve your goals"	Four types: 1) Premium (prestige) pricing 2) Penetration pricing 3) Economy pricing 4) Price skimming	[63]
	"Pricing strategy is the method of pricing a business uses to determine how much to sell their goods or services for"	Three types: 1) Cost plus pricing 2) Competitive based pricing 3) Value based pricing	[64]
	"Pricing strategy concerns the method of setting your price points in a way that establishes your product as competitive in the eyes of potential buyers"	–	[65]
	"Pricing Strategies: The way your pricing model is presented and marketed"	Eight types: 1) Positioning 2) Discounting 3) Anchoring 4) Charm Pricing 5) Bundling 6) Perks 7) Localization 8) Making a recommendation	[45]
	"[Pricing] strategy: what is the goal of the price"	Three types: 1) Skimming, 2) Maximization, 3) Penetration	[61]
	"The goal of pricing strategy is to assign a price that is the monetary equivalent of the value the customer perceives in the product while meeting profit and return on investment goals"	–	[33]
Pricing model	"Pricing Models: The method by which user(s) pay to use your product and for how much is called the pricing model"	Seven types: 1) Flat-Rate Pricing 2) Usage-Based Pricing 3) Tier-Based Pricing 4) User-Based Pricing 5) Feature-Based Pricing 6) Credit-Based Pricing 7) Hybrid Pricing	[45]
	"The pricing model is a method used by a company to determine the prices for its products or services. A company must consider factors such as the positioning of its products and services as well as production costs when setting the prices of its goods and services"	Eight types: 1) Per User Pricing 2) Per Storage Pricing 3) Per Feature Pricing 4) Freemium 5) Price Per Contact / Item / Unit or Tiered Pricing 6) Processor Time / Data Transferred (Pay as You Go) 7) Advertising 8) Broker Fee Pricing Model	[66]
Pricing structure	"Pricing structure fundamentally answers the question, "How much do I charge for my product?" by helping you determine the relationship between the value of your product or service (and especially how your customers perceive that value) and the costs incurred to create/provide it"	Seven types: 1) Singular/flat-rate pricing 2) Tiered pricing 3) Variable pricing 4) Tiered and variable 5) Per-user pricing 6) Usage-based pricing 7) Freemium	[65]
	"[Pricing] Structure: what is the pricing rubric?"	Three types: 1) Linear Pricing 2) 2 Part Tariff 3) 3 Part Tariff	[61]

Table 10
SaaS pricing challenges.

No.	SaaS pricing challenges	Ref.
1	Developing a coherent decision-making process within the company	[33,67–69]
2	The complexity of frameworks and models that should support decision-making	[70]
3	Lack of competences for performing required market, customer, and pricing analysis	[71–73]
4	Lack of proper information for the analysis	[74]
5	Market dynamics and uncertainty	[53]

Table 11
Factors affecting SaaS pricing

		WL Items	WL %	GL Items	GL %
Market	Market size and demand	8	14%	30	9%
	Market structure and competitors' positions	33	58%	44	13%
	Targeting types of customers and market segments	1	2%	34	10%
	Delivery and sales channels	7	12%	12	4%
	Dynamics of technological progress	1	2%	1	0%
Company	Business goals and opportunities	5	9%	13	4%
	Company size and maturity	1	2%	33	10%
	Ownership and financing	14	25%	0	0%
	Resources and competences availability	0	0%	8	2%
	Costs and cost structure	30	53%	56	17%
Consumers	Perceived value and willingness-to-pay	41	72%	82	25%
	Information asymmetry and customer uncertainty	2	4%	2	1%
	Customer satisfaction and loyalty	2	4%	7	2%
	Customer acquisition costs	0	0%	19	6%
	Adoption readiness	3	5%	13	4%
	Customer usage patterns	0	0%	1	0%
	Switching and migration costs	13	23%	10	3%
Product	Network effect	10	18%	8	2%
	Lifecycle stage	4	7%	27	8%
	Competitive advantage	3	5%	7	2%
	Functions and features	5	9%	77	23%
	Quality attributes	23	40%	8	2%
	Software architecture	4	7%	0	0%
	Lease and usage period	9	16%	6	2%

Only one academic paper explicitly investigated factors affecting SaaS pricing decision-making [75,76]. All other WL and GL publications only mentioned them while formalizing model assumptions, comparing different pricing strategies and models, or explaining possible lessons learned from a particular real-world case study analysis. In total, we were able to identify 24 factors that could affect the SaaS pricing strategy.

We further classified these into four categories based on the source of influence: *Market, Company, Consumers, Product*. The first two categories capture factors from the external environment, whereas the latter two categories are associated with the internal environment. These factors are presented in Table 11. As with SaaS pricing challenges, we found a gap in the coverage of these factors in WL and GL publications. Even more important, current studies do provide a quantitative or qualitative assessment of the impact of these factors on pricing performance and decision-making.

PEC #12: A wide range of SaaS pricing factors have been mentioned. However, not all factors have received much attention in academic literature, and the assessment of these factors' influence on pricing has never been systematically conducted.

RQ 3.3. What SaaS pricing decision-support frameworks, if any, have been proposed and how they vary?

We identified eleven decision-support SaaS pricing frameworks (F1-F11), one of which (F1) was proposed in WL and ten in GL. Several publications provided frameworks aimed to structure SaaS pricing and analyze that were already explored in ***RQ 2.3.*** While they might be useful in the decision-making, such frameworks do not explicitly guide product and pricing managers through the decision-making process. We did not include them in the analysis in this section. At the same time, many publications provided an extensive range of non-systematic guidelines in a narrative form or recommendations regarding certain SaaS pricing aspects (e.g., [77]). We also did not include them and considered only structured decision-making frameworks. Characteristics and structures of identified frameworks are summarized in Table 12.

Proposed frameworks vary regarding covered SaaS pricing aspects (Table 13) that might affect decision-making. Publications presenting these frameworks did not offer any evidence of their implementations in practice. The question of their joint usage also remains open.

PEC #13: No single SaaS pricing decision-making framework addressed all SaaS pricing aspects, most of them assessed decision-making on a strategic level.
PEC #14: No information on actual usage of proposed decision-support frameworks by SaaS providers has been disclosed.

5. Discussion

In this section, we summarize and discuss our research findings and provide guidelines on further research in the SaaS pricing area aimed to close any identified research gaps. We also discuss our experiences of doing an MLR study on such a multidisciplinary topic as SaaS pricing and describe the limitations of the study.

5.1. State-of-the-art and state-of-the-practice in SaaS pricing

SaaS pricing is a topic of interest in several different research domains and has been explored from different perspectives. Besides classical economic, managerial, and behavioral schools of thought that can be identified in generic pricing research [9], SaaS pricing has received attention in engineering research communities, including software

Table 12
SaaS pricing frameworks.

Code	Framework focus	Framework structure	Ref.
F1	SaaS pricing development	Three-stage framework: 1) Data collection 2) Strategy analysis 3) Strategy establishment + Customer Reaction	[78]
F2	SaaS pricing development	Six-step framework: 1) What is the customer's value of the product? 2) Is the customer aware of this value? 3) Can the customer base be segmented? 4) Is the customer's demand variable or uncertain? 5) Establish a price floor 6) What are the value metrics that are most important to the customer?	[79]
F3	SaaS pricing development	Four-step framework: 1) Define upper bound: the max value the product has for customers 2) Define lower bound: the min amount needed to charge to cover products' costs 3) Identify reasons to charge less than your maximum value 4) Structure pricing model as a compromise between upper bound and lower bound	[80]
F4	SaaS pricing development	Five-steps framework: 1) Use qualitative and quantitative data to collect customer insights 2) Quantify your website personas with data findings from qualitative and quantitative research 3) Define your SaaS key value metric 4) Create SaaS pricing tiers a) Define the minimum amount a customer must pay to cover your costs b) Define the maximum amount a customer is willing to pay for your software c) Assess competitive pricing d) Map out your pricing tiers e) Align your value metric with pricing tiers based on buyer personas 5) Design Your Pricing Page a) Define the elements of your pricing page b) Optimize your SaaS page design	[81,82]
F5	SaaS pricing development	Four-step framework: 1Industry benchmark, 2) Competitive analysis, 3) Economic value analysis, 4) Market research a) Qualitative method b) Qualitative method	[83]
F6	SaaS pricing development	Four-step framework: 1Analysis 2) Straw man (Model) 3) Testing 4) Iteration and rollout	[84]
F7	SaaS pricing development	Seven-step framework: 1) Fundamental frameworks 2) Identifying metrics 3) Price options 4) Identifying target market 5) Quantifying value of Product 6) Understanding operational costs 7) Taking into account competitors	[48]
F8	SaaS pricing development and evolution	Three-stage framework: 1) Pricing at the Seed Stage 2) Pricing at the Expansion Stage 3) Pricing at the Growth Stage	[6,85]
F9	SaaS pricing changing	Four-step framework: 1) Market and customer research 2) Reviewing price options with a customer advisory panel 3) Run an impact analysis 4) Setting up a communication plan 5) Launch pricing	[86]
F10	SaaS pricing changing	Five-steps framework: 1) Agree on your new pricing 2) Communicate internally with all teams 3) Inform your existing customers 4) Transfer your existing customers 5) Communicate your price changes	[6]
F11	SaaS pricing changing/ problem-solving	Four-step framework: 1) Defining your growth problems 2) Determining the root causes 3) Testing your solutions 4) Implementation, then iteration	[22]

Table 13
SaaS pricing aspects coverage in decision-making pricing frameworks

		F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Strategic level	Competition and market research				●	●	●		●	●		
	Customer segmentation and value proposition	●								●		
	Structure and models	●	●	●		●	●		●			
Tactical level	Versioning and bundling			●	●		●			●		
	Transparency and communication				●					●		
	Customer and usage analytics				●					●		
Operational level	Ownership and decision-making											●
	Performance measurement and evolution						●		●		●	
	Feedback, monitoring and control						●					

product management, software engineering, and information systems.

Our review revealed that, since 2012, the number of academic publications published annually has been declining, and the total number of publications directly focused on SaaS pricing has been fewer than seven per year. Possible reasons for the declining number of WL publications may be that scholars do not see SaaS as a promising field of research or do not clearly understand the practice of SaaS pricing and its challenges. Scholars may also consider that all relevant aspects of SaaS pricing have already been studied, or that general pricing literature provides sufficient answers, and there is, therefore, no need for dedicated research regarding the SaaS context. However, the difference between the number of WL and GL publications and the growing number of GL publications could indicate that much research is still missing.

These publications have employed different methods and covered various aspects of SaaS pricing. It became clear that academia has not delivered a coherent body of knowledge on SaaS pricing, and multiple research avenues remain open for further investigation. We believe that there is room for qualitative research, such as ethnography and grounded theory, because they would allow scholars to gain a first-hand, in-depth understanding of the intricate, context-specific processes of SaaS pricing, especially of their design and implementation. Also, generalization and verification of hypotheses quantitively, grounded either in

Table 14
Summary of PECs and recommendations on SaaS pricing research.

No.	PEC	Recommendations
1	The number of WL publications per year is declining, indicating a decline of interest by scholars to pursue this topic. However, the number of GL publications is growing.	• Academic research on SaaS pricing can advance pricing practices and contribute to the further sustainable development of the rapidly growing SaaS market. • Scholars need to ensure that the research aim corresponds with pricing challenges facing SaaS companies
2	There is an imbalance between the amount of GL and WL, with a low number of publications by scholars and a high number of publications by practitioners.	• Special issues and conference tracks on SaaS pricing and might provide incentives for scholars to perform such studies. • Scholars might consider non-systematic and often subjective recommendations and observations from the GL as a starting point for academic investigations
3	Absence of any cross-references between WL and GL and a scarcity in cross-references between research domains.	• Collaboration among scholars and experts in this area may promote further research and increase the value of practical implication of their studies. • SaaS industry might benefit from scholars' proactiveness in sharing research findings in the SaaS pricing area with practitioners by being involved in preparing GL publications
4	The portfolio of research methods employed in academic studies presented in WL is mostly limited to theoretical modeling and non-systematic reviews.	To advance knowledge of SaaS pricing, a broader range of studies should be conducted.
5	Both GL and WL lack quantitative studies and extensive surveys	Range of studies on SaaS pricing might include large-scale quantitative studies based on extensive surveys and collect publicly available data and companies' datasets.
6	Scholars, unlike practitioners, tend to avoid contextualization of the research, assuming homogeneity of SaaS solutions, consumers, and markets.	Research can benefit from assessing the influence of different product characteristics and specific contexts on pricing.
7	Existing academic studies provide quite a narrow view of the objective of SaaS pricing. They do not disclose the role of pricing in the organizational context. Practitioners deliver a more comprehensive, but still disorganized vision of pricing roles and objectives.	Exploring the role of SaaS pricing and its objectives is essential step in further in further SaaS pricing research
8	Research on SaaS pricing has mainly focused on general pricing issues and strategy design. There is little research on pricing tactics, organizational processes, and practices in SaaS pricing.	Scholars should conduct studies to explore SaaS pricing tactics, organizational processes, and practices. Existing generic studies on pricing tactics and priciness processes could be reconsider and replicated in the SaaS context.
9	Scholars and practitioners introduced several approaches aimed to structure and dissect SaaS pricing from several perspectives. However, proposed approaches are inconsistent with each other.	Various SaaS pricing typologies and approaches need to be systemized to deliver a coherent meta-model of SaaS pricing
10	Scholars and practitioners seem to lack a coherent vocabulary of terms associated with SaaS pricing.	Pricing strategy research and practice will benefit from unified system of concepts and terminology that allow comparison of different pricing practices across the SaaS market.
11	Little is known about the challenges facing SaaS companies while designing and implementing SaaS pricing.	Scholars should reveal SaaS pricing challenges and address them in their studies
12	A wide range of SaaS pricing factors has been mentioned. However, not all factors have received much attention in academic literature, and the assessment of these factors' influence on pricing has never been systematically conducted.	Scholars should conduct studies to assess the influence of various factors on SaaS pricing quantitatively and qualitatively.
13	No single SaaS pricing decision-making framework can address the whole SaaS pricing problem due to the complexity of factors and their interrelations.	A systematization of different decision-support SaaS pricing frameworks is needed to assist practitioners.
14	No information on actual usage of proposed decision-support frameworks by SaaS providers has been disclosed.	Scholars need to ensure that proposed frameworks could be taken into account and employed by companies and implemented in a real-world environment.

Table 15
Recommendations for further studies on SaaS pricing.

No.	Research gap	Further research directions
1	Inconsistency in SaaS pricing area of research	• Bridge the gap between product marketing-management and economics, on the one hand, and software engineering on the other.
2	Little is known about pricing practises in companies	• Conduct longitudinal and exploratory case studies. • Conduct large-scale quantitative studies based on extensive surveys and collect publicly available data and companies' datasets.
3	Research on SaaS pricing lacks contextualization	• Develop a contextual theory of SaaS pricing. • Conduct studies to compare SaaS pricing practices in different contexts.
4	There is little research on pricing tactics, organizational processes, and practices in SaaS pricing.	• Examine existing generic pricing studies and reconsider them in the SaaS context. • Conduct research to assess various pricing methods available for SaaS companies, including psychological and behavioral ones.
5	Lack of knowledge about the challenges facing SaaS companies while designing and implementing SaaS pricing.	• Study practices and processes of pricing design and implementation in SaaS companies. • Conduct studies to identify SaaS pricing challenges and provide solutions or recommendations.
6	Existing studies do not disclose the role of SaaS pricing in the organizational context and SaaS pricing objectives	• Conduct studies to explore and assess organizational aspects of SaaS pricing. • Conduct studies to assess the relevance of different pricing objectives and interdependencies between them.
7	Lack a coherent system of terms regarding SaaS pricing.	• Develop a meta-model of SaaS pricing. • Provide precise classification and definitions of pricing-related concepts and terms based on previous research and industrial practices.
8	Lack of research aimed to assess the influence and importance of various factors that affect SaaS pricing	• Conduct studies on disclosure factors affecting SaaS pricing. • Assess the influence of various factors on SaaS pricing quantitatively and qualitatively.
9	Lack of integrated SaaS pricing framework that might be used by SaaS companies in	• Conduct a comparative analysis of existing frameworks from different perspectives. • Employ design science and action research to deliver a decision-support model and tool.

theoretical models or findings of qualitative studies, would be beneficial for the field.

The current research appears fragmented and separated from practice. We found a wide variety of notions, terms, and concepts proposed in publications. However, the presented frameworks and models designed to support decision-making in SaaS pricing seemed to lack coherency. It was hard to find evidence that research findings are actually used by practitioners, and researchers seemed not to expect that to happen because they seldom made claims that their findings would be directly usable in the real world. The public landscape of GL confirmed the presence of promising niches for further quantitative and design science research.

The analysis of the GL publications indicated how SaaS pricing is essential for the industry. These publications have delivered a broad range of recommendations and observations on SaaS pricing. While the GL publications were less systematic, they still covered a broader range of SaaS pricing aspects. Unlike practitioners, scholars can offer more systematic and rigorously developed solutions and recommendations to support SaaS companies in their pricing.

By combining WL and GL publications' findings, we classified SaaS pricing aspects, objectives, affecting factors, and challenges facing SaaS providers. Both academic research and industry observation showed that there is no unified approach to define and implement pricing. The findings of the conducted MLR have been summarized to emphasize the major research themes and practical challenges of SaaS pricing practices' transformation. The PECs and recommendations of this study are summarized in Table 14. The next subsection provides guidelines for further research in this area.

5.2. Further research directions

Identified PECs allowed us to go further and explicitly specify research gaps, as well as provide recommendations for further research as summarized in Table 15. The provided research directions highlight promising research avenues, following which will fill the theory–practice gap and overcome the inconsistency in research on SaaS pricing.

5.3. Importance of GL in the study

MLR is gaining momentum in academic literature, especially in areas that are critical for both scholars and practitioners when there is a need for interdisciplinary investigations and different perspectives. MLR combines state-of-the-art research and state-of-the-practice expertise when there is a clear gap between academic literature and actual practice [10,87]. While MLR methodology has been widely used in medicine and education sciences, researchers in management and engineering recognized its value less than a decade ago [9,88]. To the best of our knowledge, this MLR on SaaS pricing is the first of its kind, not just in the area of SaaS pricing, but also in broader fields such as software product management, pricing management, and cloud computing in general.

This study began as a regular, systematic literature review intended to identify the current state-of-the-art on SaaS pricing. After the body of academic literature was collected, it became clear that the research on SaaS pricing is fragmented and inconsistent. It also became evident that the findings of existing studies would be of little use to real SaaS companies, and the research is separated from the extensive body of practical expertise. It convinced us to employ a multivocal methodology and assess not just the current state-of-the-art, but to match it with the state-of-the-practice in this area by conducting a literature review on both WL and GL. Below, we enumerate the ways, including GL, that enabled the identification of emerging research topics and provided a clear picture of the current body of knowledge on SaaS pricing. First, the MLR approach allowed us to identify gaps in the existing academic body of literature. For instance, we found that SaaS pricing tactics and organizational issues have not received proper attention among scholars, while they seem to be relevant to practitioners based on the GL analysis. Facing a lack of dedicated studies, practitioners have had to investigate general research on pricing tactics and organizational issues and intuitively conclude which research findings might be valid to the SaaS domain. Further research might close this gap.

Second, the MLR approach enabled us to provide a comprehensive taxonomy for different SaaS pricing-related aspects (i.e., objectives, factors affecting, methods available). Including GL made the range of opinions and pieces of evidence broader and allowed us to extend initial taxonomies based on WL. Evaluation of the importance and significance of different items within these taxonomies required additional research,

but it revealed the variety fully and helped us to meet the exploratory aim of this study. This would not have been possible without a thorough investigation of GL.

Third, GL brought contextualization to the SaaS pricing research. Most WL publications considered SaaS solutions as a homogeneous population without exploring the various products, markets, and consumer characteristics. GL publications showed that efficient SaaS pricing decision-making requires a deep understanding of the context. This led us to call for further contextualization of the research on SaaS pricing.

Finally, analytical methods and questions raised in GL publications have revealed several directions for future research. For example, we identified several GL publications with recommendations based on observing pricing pages of a large number of SaaS companies. While practitioners' publications did not go further than basic descriptive statistics, their findings suggested that a sufficiently better understanding of pricing practices and factors affecting these practices could be delivered by performing an econometric analysis on similarly collected data.

5.4. Challenges and limitations of the study

MLR is gaining momentum as a research methodology for managerial and engineering research, including software engineering and software product management. Several publications (i.e., [9,88]) proposed guidelines and recommendations for the research process. Still, we faced many challenges in the implementation of the study. Below we list these limitations and challenges and explain how we took precautions to minimize them.

First, finding relevant GL publications was more challenging than collecting WL from databases. Many publications created by practitioners are not public and searchable. While we were able to identify several publications by leading global consulting companies (e.g., PWC, McKinsey) or software giants (e.g., Amazon Web Services), much content was produced by individual experts or SaaS companies. Leading strategic consulting companies (e.g., BCG or Accenture) and other software giants (e.g., Apple and Google) have their own frameworks and approaches to pricing that could be useful for other SaaS companies. However, their approaches to pricing are not published and were not included in the scope of literature. Yet, the number of included GL publications proved to be sufficient to obtain a clear picture of the current state-of-the-practice.

Second, during the process of data extraction, we found that not many papers specified sufficient details to fulfill all required table fields. This might be quite typical for systematic literature reviews, but including GL, further complicated this issue. GL publications tend to be focused on conclusions and recommendations without providing a chain of evidence that led to these findings. In many cases, we had to complete the comparison table based on clues that were not completely clear, mainly when we extracted SaaS pricing aspects, objectives, and pricing affecting factors. Therefore, there may be some small inaccuracies in the inferred data, but we do not consider this as a threat to the validity of the findings. These potential inaccuracies cannot significantly change the whole picture.

Third, we considered only publications where it was clearly stated that the focus of the study or topic of the publication was the pricing of SaaS. As for WL, we did include papers from the economic and business domains that did not use the term SaaS, but still clearly expressed that their research subject met the SaaS definition. Nevertheless, we did not include papers on SaaS without a clear focus on pricing issues or papers on software pricing in general. The same criteria were applied to GL publications. We also did not include short blogs and social media posts. We remained confident that the amount of GL and WL included would be sufficient to deliver results that disclosed the state-of-the-art and the state-of-the-practice on SaaS pricing.

Finally, this research was carried out by two co-researchers. To ensure bias avoidance and the validity of the research procedure, segregation of responsibilities was established between the two co-researchers. The first author was responsible for performing activities determined by the research protocol at all stages of the research, while the second author carried out monitoring/reviewing of the study at each stage, following a defined supervisory strategy. All collected information was publicly available for external auditing. Additionally, the article itself and its findings were further discussed with industry experts specializing in software product management and pricing.

6. Conclusions

Migration to the SaaS business model has pointed out the importance of pricing in ensuring software companies' success. SaaS pricing also impacts many processes inside a company, affects different business units, and requires sophisticated analysis. While understanding the complexity of pricing and its strategic role, many companies find themselves incapable of performing proper pricing and rely on a trial-and-error decision-making approach. The academic community has failed to equip the industry with trustworthy pricing approaches, frameworks, and guidelines. We also lack a systematic investigation of existing SaaS pricing practices that vary significantly across the industry.

To the best of our knowledge, this research is the first to compare state-of-the-art and state-of-the-practice of SaaS pricing. Using the MLR formal procedure, we collected **387** items of WL and GL that directly addressed SaaS pricing issues. We extracted information from the collected literature and synthesized it to provide answers to the questions of what the current status of academic research and practical expertise on SaaS pricing is (*RQ1*), how SaaS pricing is defined and disseminated in academic literature and considered by practitioners (*RQ2*), and in which way existing studies support SaaS providers in designing and implementing pricing (*RQ3*). While the primary aim was to raise the awareness of SaaS pricing issues and stimulate further discussion, we believe our analysis brought some clarity to the only just emerging body of knowledge on SaaS pricing.

We recognized gaps in the current research on SaaS pricing. These identified gaps are mostly related to the inconsistency among different perspectives, the limited scope of methodologies employed, and the lack of clarity on the real obstacles of designing and implementing pricing in SaaS companies. The analysis of WL and GL publications allowed us to propose a research agenda in the form of identified research gaps and recommendations for further studies as presented in Table 14 and Table 15.

Our further work will be aligned with the proposed recommendations and can be unfolded in two directions. First, we intend to perform a large-scale quantitative study on SaaS pricing practices in the software industry with available public information. This study will complement a survey among product managers in SaaS companies responsible for developing and implementing a pricing strategy. We plan to analyze the decision-making process, challenges faced, as well as models and methods used. Second, we aim to use a constructive and design science approach to develop an integrated decision-making framework that SaaS companies can use to establish pricing processes and practices. We intend to test the framework in SaaS companies further.

We also gained significant insights while conducting this MLR study. This methodology's enduring popularity is justified and understandable as it aims to identify new sources of evidence, knowledge, and expertise to the research in highly industrial relevant areas. However, including GL makes the research process more complicated and adds certain threats to validity. Within this paper, we have reported on the challenges that we faced while conducting our study on SaaS pricing and the steps taken to overcome them. This will hopefully help other scholars to avoid these challenges.

Declaration of Competing Interest

No.

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